INCH-POUND
MIL-DTL-32813
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DETAIL SPECIFICATION

TRANSFORMER, DRY-TYPE, POWER, DISTRIBUTION, MEDIUM VOLTAGE, THREE-PHASE, 60-HERTZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification is applicable to 60-hertz, three-phase, dry-type, medium voltage distribution and power transformers (see 6.3.10) for shipboard use.

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-208	-	Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces)
A-A-59486	-	Padlock Set (Individually Keyed or Keyed Alike)
A-A-59487	-	Padlock (Key Operated)
A-A-59781	-	Light Emitting Diodes for Use as Indicator Lights

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-917	-	Electric Power Equipment, Basic Requirements for
MIL-DTL-2036	-	Enclosures for Electric and Electronic Equipment, Naval Shipboard
MIL-DTL-19836	-	Coolers, Fluid, Industrial, Air, Motor and Generator, Naval Shipboard
MIL-DTL-22577	-	Heating Elements, Electrical: Cartridge, Strip, and Tubular Type
MIL-T-24388	-	Thermocouple and Resistance Temperature Detector Assemblies, General Specification for (Naval Shipboard)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

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MIL-DTL-32483 - Switchgear, Power, Hard-Mounted, Medium Voltage, Naval Shipboard

MIL-DTL-32656 - Arc Fault Protection System, General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

MIL-STD-202-103 - Method 103, Humidity (Steady State)

MIL-STD-202-301 - Method 301, Dielectric Withstanding Voltage

MIL-STD-202-302 - Method 302, Insulation Resistance

MIL-STD-461 - Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment

MIL-STD-740-2 - Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment

MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests

MIL-STD-1310 - Shipboard Bonding, Grounding, and Other Techniques for Electromagnetic Compatibility, Electromagnetic Pulse (EMP) Mitigation, and Safety

MIL-STD-1399-300-1 - Interface Standard, Section 300, Part 1, Low Voltage Electric Power, Alternating Current

MIL-STD-1399-300-2 - Interface Standard, Section 300, Part 2, Medium Voltage Electric Power, Alternating Current

DOD-STD-1399-301 - Interface Standard for Shipboard Systems, Section 301A, Ship Motion and Attitude

MIL-STD-1399-407 - Interface Standard, Section 407, DC Magnetic Field Environment

MIL-STD-1474 - Noise Limits

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-470 - Designing and Developing Maintainable Products and Systems, Volume I

(Copies of these publications are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9510-AB-ATM-010 - Nuclear Powered Submarine Atmosphere Control Manual; Volume 1

(Copies of the chapter titled “Material Control Program” are available by email request to CommandStandards@navy.mil.)

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T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this publication are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To request an MBPS account, submit a ticket to the ServiceNow Assistance Portal (SNAP), https://snap.navair.navy.mil/mbps (DoD PKI Certificate required).)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS411-1 - Hazardous Materials Target List

(Copies of this publication are available online at www.aia-aerospace.org.)

IEEE

IEEE C37.20.2 - IEEE Standard for Metal Clad Switchgear

IEEE C37.20.7 - IEEE Guide for Testing Switchgear Rated Up to 52 kV for Internal Arcing Faults

IEEE C57.12.01 - IEEE Standard for General Requirements for Dry-Type Distribution and Power Transformers

IEEE C57.12.91 - IEEE Standard Test Code for Dry-Type Distribution and Power Transformers

IEEE C57.134 - IEEE Guide for Determination of Hottest-Spot Temperature in Dry-Type Transformers

(Copies of these publications are available online at www.ieee.org.)

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 60076-18 - Power Transformers – Part 18: Measurement of Frequency Response

(Copies of this publication are available online at www.iec.ch.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025 - General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this publication are available online at www.iso.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 General requirements. Unless otherwise specified (see 6.2), the medium voltage transformer shall comply with the applicable requirements of IEEE C57.12.01.

3.3 Safety. Unless otherwise specified (see 6.2), safety features shall be in accordance with MIL-DTL-917.

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3.3.1 Accidental contact. When energized during normal operations, the transformer shall provide protection to personnel against accidental contact with voltages in excess of 30 volts root mean square (rms) or direct current (DC) with all removable panels in place and doors closed.

3.4 Construction requirements.

3.4.1 Materials and coatings. Unless otherwise specified (see 6.2), materials and coatings contained in or on the transformer shall meet the requirements of MIL-DTL-917.

3.4.1.1 Non-preferred component materials. Non-preferred materials contained in or on the transformer shall be in accordance with MIL-DTL-917 and shall only be used if approved by the Technical Authority (see 6.3.9) prior to the start of first article testing, as specified (see 6.2).

3.4.1.2 Core and winding material. Core laminations shall be electrical grade steel. Coils shall be wound of copper magnet wire in accordance with MIL-DTL-917.

3.4.1.3 Coatings.

3.4.1.3.1 Painting and powder coating. Transformer enclosures shall be painted or powder coated in accordance with MIL-DTL-917.

3.4.1.3.2 Non-preferred component coatings. Corrosion-susceptible coatings not in accordance with MIL-DTL-917 are non-preferred and shall only be used if approved by the Technical Authority prior to the start of first article testing, as specified (see 6.2).

3.4.1.3.3 Conformal coating. All circuit boards shall be conformal coated in accordance with MIL-DTL-917.

3.4.1.4 Insulation.

3.4.1.4.1 Insulating material. Insulating material for components other than windings shall be a minimum of class B (266 °F [130 °C]) in accordance with MIL-DTL-917. Insulating material for windings shall be a minimum of class F (311 °F [155 °C]) in accordance with MIL-DTL-917. The coil for each transformer shall have resin applied by the vacuum pressure impregnation process specified in MIL-DTL-917. The core for each transformer shall have resin or varnish applied by the vacuum pressure impregnation process or the dipping process specified in MIL-DTL-917. Solid cast windings may also be used.

3.4.1.4.2 Insulating material sealing. Insulating material that is susceptible to absorption of moisture shall be sealed to prevent the absorption of water. Insulating material that is susceptible to absorption of moisture when cut edges are present shall have cut edges sealed to prevent the absorption of water.

3.4.1.4.3 Insulation flame resistance. Insulating materials shall be flame-resistant in accordance with MIL-DTL-917. Insulating material not in accordance with MIL-DTL-917 requirements may be used, but such material shall be subjected to the flame testing of 4.6.23.1.

3.4.1.4.4 Insulation coverage. Winding conductors and connections shall be covered with insulating material throughout except at terminal connection points. Connection points at terminals shall be covered by removable boots.

3.4.2 Fasteners. Unless otherwise specified in the individual equipment specifications, fasteners and locking devices shall be provided in accordance with MIL-DTL-917 with the exception that bus bar, cable fasteners, and locking devices shall be in accordance with MIL-DTL-32483. A visual method to check bus bar and cable fastener tightness (such as torque check paint) shall be provided.

3.4.3 Wiring and cabling.

3.4.3.1 Control and instrument cable. Control and instrument wire and cable, wire and cable harnessing, hook-up wire, terminations, and markings shall be in accordance with MIL-DTL-917, unless otherwise approved by the Technical Authority.

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3.4.3.2 Power cable connections. Connections to the ship's power cables shall be provided using two-hole lugs. The ship's power cable entrance for primary and secondary winding (see 6.3.4 and 6.3.8) terminations, power cable type and quantity, and terminal lug shall be as specified (see 6.2). Single-hole lugs shall be permitted for cable shield connections.

3.4.3.3 Auxiliary compartment connections. Low voltage controls, fans, indicators, instrument transformers, Resistance Temperature Detectors (RTD), and anti-condensation heater interfaces shall all be wired to a single low voltage auxiliary compartment physically isolated from the power components and shall be easily accessible. Access to the low voltage auxiliary cabinet shall be via covers.

3.4.3.4 Low voltage wiring. Low voltage wiring shall terminate on terminal boards mounted in the low voltage auxiliary compartment. Terminal boards shall provide spare capacity in accordance with MIL-DTL-917. The terminal board control wiring shall interface with the ship's wiring via externally mounted multi-pin connectors mounted on the exterior of the enclosure. Each individual signal interface shall be designed to accommodate a shield from the plug connectors to the terminal boards. Power for heater circuits and instrument transformer wiring shall be separated from RTD and signal wiring.

3.4.4 Buses and bus supports. All buses shall be able to provide full rated current at a 122 °F (50 °C) ambient temperature. The bus bar shall be rectangular copper with rounded edges, hard tempered, and silver plated in accordance with MIL-DTL-32483. The bus bar size shall meet the design requirements of MIL-DTL-32483. Bus work shall be supported to withstand the maximum fault current levels.

3.4.4.1 Bus bar insulation. The bus shall be fully insulated in accordance with IEEE C37.20.2. Bus bar connection points not covered by insulation shall be covered by removable boots (see 4.6.23).

3.4.5 Enclosure. Unless otherwise specified (see 6.2), enclosures shall be in accordance with MIL-DTL-2036. Enclosures shall provide for the entrance of the ship's cables. Accessibility shall be provided with clearance for installation and maintenance or electrical connections.

3.4.5.1 Lifting means. The transformer shall be provided with eyebolts or an equivalent means for lifting and handling. If the attachments for lifting extend beyond the transformer outline, they shall be removable. Degree of enclosure (e.g., drip-proof) integrity shall be maintained with any lifting attachments removed.

3.4.5.2 Enclosure access panels or covers. Unless otherwise specified (see 6.2), enclosure access panels and covers shall be in accordance with MIL-DTL-2036. Hinged and removable panels on the front and rear of the unit shall be secured by thumb screws with slots for the use of a screwdriver.

3.4.5.2.1 Enclosure security. The transformer shall have no externally accessible data ports or wireless connections. All unused or unnecessary ports inside enclosures shall be capable of being disabled physically or logically. Physical access to the ports shall require opening enclosure access panels and covers. Any data ports that will allow for the modification of existing software or installation of new software on the device shall be located within a secured enclosure. When provided, all enclosures with data ports shall have access panels or covers and shall utilize lead-wire type or other tamper-evident seals and padlocks in accordance with A-A-59486 or A-A-59487. Access panels or covers for electrical connections shall include an attached printed connection diagram of associated terminal connections.

3.4.5.3 Neutral bar. The transformer shall contain a neutral bar for termination of the cable stress cones and secondary shields. The requirements for the neutral bar shall be specified by the vendor (e.g., hole size/spacing, hole quantity, minimum bus bar thickness, or minimum isolation level). Provisions shall be added such that the bus can be grounded if necessary and included in the cabinet at shipment.

3.4.6 Marking transformer buses. Buses shall be clearly marked in a visible location with their polarity or phase identification. If marking is accomplished by painting (stencil) application, the ink shall conform to A-A-208, type III.

3.4.7 Phase identification. Phase identification of A-C buses shall be indicated by the capital letters A, B, or C, as applicable. A neutral bus of a four-wire, three-phase circuit, where used, shall be indicated by the capital letter N. Letters A, B, and C, in that order, shall indicate the phase sequence.

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3.4.8 Bus bars. Bus bars facing and closest to the rear of the unit shall be stenciled "DANGER _____ VOLTS" with applicable voltage entered. Letters shall be white and not less than ¾ inch high. As an alternative to stenciling, bus bars may be provided with a warning plate bearing the same legend. Letters shall be not less than ¾ inch high. The plate shall be pressure-sensitive, adhesive-backed elastomeric film with a carrier or separator sheet applied over the adhesive. Film shall consist of a white printed opaque polyvinyl fluoride film 0.002 inch thick laminated to a 0.001-inch clear polyethylene terephthalate film. Plate adhesive shall consist of an acrylic base vinyl polymer of a pressure sensitive type. A minimum of two warning stencils or plates per unit shall be provided, one in the upper section and one in the lower section.

3.5 Space heater. The transformer shall be equipped with a combination off/automatic, switch selectable, strip-type space heater for humidity control in accordance with MIL-DTL-22577. The heater, in the automatic mode, shall only be allowed to be energized when the power transformer is de-energized. A white light shall come on to indicate heater power is available (regardless of power transformer status), and another white light shall come on to indicate when the heater is energized. When the transformer is energized, additional heat is not required, and the heater shall be automatically secured when the control switch is in the automatic position or shall be secured manually. When specified (see 6.2), the heater shall be thermostatically controlled (fixed or adjustable) to prevent damage to components due to excessive heating (see 4.6.10.2).

3.5.1 Space heater power. The space heater shall be powered from a separate low voltage source (see 3.9.1). The ship's heater power interface shall be via a multi-pin plug. The plug shall be wired to a terminal board located in the auxiliary compartment. Additionally, each of the energized leads of the heater circuit shall be fused. The fuses shall be installed on the outside face of the auxiliary compartment cover. The fuse shall be sized to protect the circuit yet allow the heaters to operate normally.

3.6 Identification plates, information plates, and marking. Unless otherwise specified (see 6.2), identification plates, information plates, and markings shall be in accordance with MIL-DTL-917.

3.6.1 Identification plate information. Identification plates shall indicate the following information at a minimum:

a. The manufacturer's name and model
b. The manufacturer's serial number (or identification mark)
c. The year of manufacture
d. The number of phases
e. The rated power in kilovolt-amperes (kVA) (see 6.3.5)
f. The rated frequency
g. The rated no-load voltage in primary and secondary sides
h. The rated current in primary and secondary sides
i. Permissible temperature limits
j. The weight
k. The National Stock Number
l. Class insulation
m. Connection diagram
n. Available primary tap (see 6.3.11) settings (if applicable)
o. Available secondary tap settings (if applicable)
p. Phasor diagram (polyphase transformers)
q. Percent impedance
r. Percent regulation (see 6.3.7) at 0.8 power factor
s. Percent regulation at 1.0 power factor

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3.7 Temperature.

3.7.1 Ambient temperature. The ambient operating temperature range shall be from 32 to 122 °F (0 to 50 °C). The non-operating ambient temperature range shall be from -40 to +167 °F (-40 to +75 °C).

3.7.2 Temperature monitoring. Winding hotspot temperature shall be monitored by RTDs placed in each phase winding located at the hottest spot as determined by IEEE C57.134. Wiring from all RTDs shall run in a dedicated wire way or conduit to a terminal box with access for connecting them to the monitoring system without exposure to energized parts. All RTDs shall be provided with a backup RTD. Wires and terminations shall be permanently labeled as to their purpose, polarity, and whether normal or backup. RTDs shall be in accordance with MIL-T-24388. Additional locations of RTDs may be required, as specified (see 6.2). Embedded RTDs shall be insulated to withstand contact with energized windings for the lifetime of the transformer.

3.7.2.1 Temperature warning and shutdown sensor. When specified (see 6.2), RTDs placed in each phase winding shall provide winding hotspot temperature signals to allow shutdown of the transformer or provide warning signals at prescribed insulation temperature ratings to protect the transformer.

3.8 Temperature limits.

3.8.1 Temperature measurement. Primary and secondary winding temperature rises shall be measured by the resistance method of IEEE C57.12.91 in accordance with table I. Unless otherwise specified (see 6.2), core temperature rises shall be measured by the thermometer method of MIL-DTL-917 in accordance with table I. As an alternative, when specified (see 6.2), core and winding temperatures shall be measured by the embedded detector method of MIL-DTL-917. Winding temperature using the transformer's embedded RTDs shall be less than the rating of the applicable insulation class.

TABLE I. Maximum temperature rises of windings and cores.

Insulation system temperature class, °F (°C)	Average winding temperature rise by resistance method, °F (°C)	Core temperature rise by thermometer method, °F (°C)
311 (155)	153 (85)	189 (105)
356 (180)	189 (105)	234 (130)
392 (200)	216 (120)	270 (150)
428 (220)	252 (140)	306 (170)

3.9 Ratings. Transformer ratings shall be as specified (see 6.2).

3.9.1 Standard power interfaces. Unless otherwise specified (see 6.2), all low voltage power interfaces shall be in accordance with MIL-STD-1399-300-1 and medium voltage power interfaces shall be in accordance with MIL-STD-1399-300-2.

3.9.2 Phase. The transformer shall be three-phase in accordance with IEEE C57.12.01.

3.9.3 KVA capacity. The transformer shall be rated for continuous load with the assembly operating at the maximum ambient temperature in accordance with IEEE C57.12.01 at the level specified (see 6.2).

3.9.4 Duty. Unless otherwise specified (see 6.2), the transformer shall be designed for continuous duty.

3.9.5 Transformer impedance. The transformer impedance shall meet the requirements in IEEE C57.12.01 and shall be as specified (see 6.2). Impedance tolerance shall be in accordance with IEEE C57.12.01.

3.9.6 Ratio. The ratio (see 6.3.6) shall be in accordance with IEEE C57.12.01 at the level specified (see 6.2).

3.9.7 Polarity and phase relation. The polarity and phase relation shall be in accordance with IEEE C57.12.01, as specified (see 6.2).

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3.9.8 Short circuit. Unless otherwise specified (see 6.2), the transformer shall meet the short circuit rating in accordance with IEEE C57.12.01.

3.9.9 Weight and volume envelope. The transformer shall not exceed the weight and volume envelope rating, as specified (see 6.2).

3.9.10 Excitation (no-load) losses. Unless otherwise specified (see 6.2), no-load losses shall be determined using IEEE C57.12.01 as reference.

3.9.11 Load losses and impedance voltage. Unless otherwise specified (see 6.2), load losses and impedance voltage shall be determined using IEEE C57.12.91 as reference.

3.9.12 Winding resistance. Winding resistance shall be measured and corrected based on the temperature with a specified tolerance recommended by the vendor in accordance with IEEE C57.12.91.

3.9.13 Schematic. The general schematic of the transformer shall be as specified (see 6.2).

3.9.14 Inrush current. The transformer shall withstand the exciting current (see 6.3.2) value of the rated multiples of full load current, as specified (see 6.2). When specified (see 6.2), a soft start circuit shall be used to limit the inrush current to the transformer.

3.9.15 Taps. When required (see 6.2), taps shall be in accordance with IEEE C57.12.01. The rating of any tap shall be as specified (see 6.2). Taps shall be no-load type (i.e., online tap changing is not required). The tap jumpers shall be of sufficient length to be field changeable (i.e., any tap configuration can be set in the ship with the existing [supplied] jumpers).

3.10 Shock. The transformer shall be in accordance with the grade A shock requirements of MIL-DTL-901. The transformer shall be hard mounted or use shock isolation devices.

3.11 Vibration. The transformer shall be in accordance with MIL-STD-167-1 for type I vibration.

3.12 General cooling. The transformer shall be self-cooled using convection air movement to meet the operating temperature requirements. Fan cooling or water-to-air heat exchangers shall not be used unless approved by the Technical Authority.

3.12.1 Fans. When specified (see 6.2), fans shall meet the following requirements:

a. Easily accessible, washable air filters shall be provided for external air intake fans.

b. Fans shall be able to be disabled by operation of an appropriate maintenance switch.

c. When fans are installed, airborne and structureborne noise testing in accordance with MIL-STD-1474 and MIL-STD-740-2, respectively, shall be included in the first article test.

3.12.1.1 Fan monitoring system. A fan monitoring system shall be provided. The system shall have features for fan failure detection and alarms.

3.12.1.2 Fan power. Fan power and control shall be via the auxiliary compartment. All energized legs of the fan power lines shall be fused.

3.12.2 Water-cooled systems. When specified (see 6.2), water-cooled systems may be utilized. Fan- and water-cooled systems may exist and operate at the same time. When water-cooled systems are utilized, the requirements of 3.12.2.1 through 3.12.2.3 shall apply.

3.12.2.1 Water leaks. Transformers with water cooling shall have a water leak sensor with alarms. The leak sensor shall be a liquid level switch/sensor with normally open operation. The transformer design shall ensure the water system is dielectrically isolated from electric power so that a leak cannot cause a dielectric failure.

3.12.2.2 Water flow and water pressure. Input water flow and water pressure characteristics to the cooling system shall be as specified (see 6.2).

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3.12.2.3 Heat exchangers. When required (see 6.2), heat exchangers shall be in accordance with MIL-DTL-19836. Heat exchangers shall form part of the water-air-cooled enclosures built in accordance with MIL-DTL-2036. Heat exchangers shall be class 4, submarine and surface ship freshwater-cooled in accordance with MIL-DTL-19836.

3.13 Noise. The transformer assembly shall meet the airborne and structureborne noise requirements of MIL-STD-1474 and MIL-STD-740-2, respectively. Acceptable levels shall be as specified (see 6.2).

3.14 Indicator lights. Indicator lights shall be LED type. LED indicator lights shall be in accordance with A-A-59781. Indicator lights shall be furnished with lamps.

3.15 Creepage and clearance.

3.15.1 Low voltage creepage and clearance. Creepage and clearance for non-insulated components shall be in accordance with MIL-DTL-917 for voltages less than 1,000 volts alternating current (VAC). Low voltage components not covered by other specifications shall meet creepage and clearance requirements in accordance with MIL-DTL-917 with set C and enclosed spacing.

3.15.2 Medium voltage creepage and clearance. Navy medium voltage is defined by MIL-STD-1399-300-2. Phase-to-phase air clearances and phase-to-ground air clearances between non-insulated energized parts shall be not less than the minimum specified in table II. Creepage distances between non-insulated energized parts and grounded metal parts shall be not less than the minimum specified in table II. Insulation on bus bars does not negate this spacing requirement. Components with design air gaps (seams) such as bus boots shall use medium voltage creepage and clearance values in the area of the air gaps. Boots on ground studs shall be treated as non-insulated.

TABLE II. Medium voltage creepage and clearance.

Voltage defined by MIL-STD-1399-300-2	Clearance, inches (centimeters)	Creepage, inches (centimeters)
1.0 to <3.3 kV (BIL: 40 kV)	2.4 (6.0)	1/
3.3 to <6.6 kV (BIL: 60 kV)	3.6 (9.1)	4.2 (10.7)
6.6 to <11 kV (BIL: 75 kV)	4.8 (12.2)	7.0 (17.8)
11 to <15 kV (BIL: 95 kV)	6.0 (15.2)	9.5 (24.1)

FOOTNOTE:

1/ This value shall be recommended by the vendor for approval by the Technical Authority.

NOTE: Abbreviations used above include Basic Insulation Level (BIL) and kilovolt (kV)

3.15.2.1 Creepage and clearance distances applied to fully insulated components. Safe insulation distances for insulated medium voltage components shall be validated by successful performance when subjected to dielectric voltage testing, lightning impulse withstand voltage (BIL) testing, and partial discharge testing in accordance with 4.6.8 and 4.6.9.

3.16 Electromagnetic interference (EMI). Unless otherwise specified (see 6.2), the transformer with electronic devices shall be in accordance with the EMI limits and criteria of MIL-STD-461 for installation both below deck on surface ships as well as installation aboard submarines.

3.17 Humidity. The transformer shall not show any detrimental effects or any accumulation of liquid water within the enclosure when subjected to the humidity methodology of MIL-STD-202-103.

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3.18 Inclined operation. Unless otherwise specified (see 6.2), the transformer shall meet the interface requirements and constraints of DOD-STD-1399-301 using the design limits for surface ships with sea state 8 or submarines.

3.19 Grounding. All external parts capable of electrical conduction shall be bonded and at the same electrical potential at all times. Provisions shall be made to allow shipboard bonding to ground potential in accordance with MIL-STD-1310.

3.19.1 Medium voltage grounding. When access panels are provided for medium voltage components, manual grounding clusters shall be installed on dedicated accessible grounding studs with removable boots using "hot sticks." In addition to the phase conductor grounding studs, there shall be a ground stud attached to the enclosure and frame ground plane. The ground stud connected to the ground plane shall be located within the enclosure perimeter.

3.20 Arc-resistant design (pressure relief). When required (see 6.2), pressure relief devices shall be present in the design based on studies of the fault levels that will occur at locations in the transformer and the permissible venting locations.

3.21 DC magnetic field test. The transformer with electronic devices shall be operational when subjected to a DC magnetic field test in accordance with MIL-STD-1399-407.

3.22 Maintainability. When specified (see 6.2), the transformer shall not require servicing by the ship's crew other than for cleaning and routine maintenance. If included, the following components may also be replaced as needed for routine maintenance: leak detector components, current transformer/potential transformer, any soft-start contactor components, fans, fuses, and filters.

3.23 Dielectric tests. The transformer design shall be capable of withstanding dielectric tests in accordance with IEEE C57.12.91 at the levels prescribed by MIL-STD-1399-300-1 and MIL-STD-1399-300-2.

3.24 Insulation resistance. The transformer design shall be capable of withstanding insulation resistance testing in accordance with IEEE C57.12.91.

3.25 Degree of enclosure. Unless otherwise specified (see 6.2), transformer enclosures shall be drip-proof (15 degrees), in accordance with class 1 of MIL-DTL-2036 for electric equipment and with MIL-STD-108.

3.26 Faraday shield (electrostatic shield). A Faraday shield shall be provided between the transformer primary and secondary windings to filter high frequency common mode voltage transients and to prevent electrostatically coupled over voltages from primary windings to secondary windings. The Faraday shield shall be grounded.

3.27 Core grounds. The transformer core shall be grounded to the transformer frame and enclosure. The grounding mechanism shall be installed such that it can be easily disconnected after shipboard installation to support testing.

3.28 Thermal imaging windows. When specified (see 6.2), thermal imaging windows shall be provided with access to view selected current carrying joints or expected hot spots in accordance with MIL-DTL-32483. The locations in the transformer required for viewing shall be as specified (see 6.2).

3.29 Arc fault sensors. When specified (see 6.2), arc fault continuous thermal monitoring devices (CTMD) and photo sensors in accordance with MIL-DTL-32656 shall be required. Photo sensors shall be capable of monitoring arc flash events at windings' terminal cable connections. Arc fault CTMD and photo sensors and any required connection components shall be installed per the arc fault detector vendor's instructions.

3.30 Fuses. Low voltage fuses shall meet the requirements of MIL-DTL-917. When required, as specified (see 6.2), medium voltage fuses shall meet the requirements of MIL-DTL-32483.

3.31 Measurement of frequency response. The frequency response measurement of the transformer shall be in accordance with IEC 60076-18.

3.32 Toxicity. When evaluated in accordance with 4.5.2, the transformer shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.5.2 and 6.4).

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3.33 Prohibited materials. The transformer shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

3.34 Off-gassing. The transformer shall be evaluated for off-gassing in accordance with the requirements of 4.7. Based on the circumstances of use and the chemical nature of the transformer, the Navy will determine whether off-gas testing is required or if an administrative assessment is acceptable. In order to be considered acceptable for use in submarines, the transformer shall be assigned to either the “Permitted” or “Limited” category (see 4.7 and 6.5).

3.35 Partial discharge. The transformer shall be capable of withstanding the partial discharge testing of 4.6.9.

3.36 Voltage regulation. The voltage regulation of the transformer shall be in accordance with IEEE C57.12.91, as specified (see 6.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection of a single medium voltage transformer sample shall be required (see 3.1). The inspection shall include the examination of 4.5 and the tests of 4.6.2 through 4.7 (see table III).

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TABLE III. Examinations and tests.

Test requirement	Requirement	Verification	First article	Conformance
General examination	3.2	4.5.1	X	X
Safety	3.3, 3.3.1	4.5.1	X 1/	X 1/
Construction requirements	3.4 to 3.4.5.3	4.5.1	X 1/	X 1/
Space heater	3.5, 3.5.1	4.5.1	X	X
Nameplates	3.6, 3.6.1, 3.4.6 to 3.4.8	4.5.1	X	X
Creepage and clearance	3.15, 3.15.2, 3.15.2.1, 3.15.2.1	4.5.1	X 1/	X 1/
Physical	3.9.9	4.5.1	X	X 1/
Indicator lights	3.14	4.5.1	X	X
Fans	3.12, 3.12.1, 3.12.1.1, 3.12.1.2	4.5.1	X	X 1/
Water-cooled systems	3.12, 3.12.2, 3.12.2.1 to 3.12.2.3	4.5.1	X	X 1/
Grounding	3.19, 3.19.1, 3.27	4.5.1	X	X
Faraday shield (electrostatic shield)	3.26	4.5.1	X	X 1/
Taps	3.9.15	4.5.1	X	X
Arc fault sensors	3.29	4.5.1	X	X 1/
Fuses	3.30	4.5.1	X	X 1/
Schematic	3.9.13	4.5.1	X	X 1/
Maintainability	3.22	4.6.2	X	--
Resistance	3.9.12	4.6.3	X	X
Polarity and phase-relation	3.9.7	4.6.4	X	X
Ratio	3.9.6	4.6.5	X	X
Excitation losses	3.9.10	4.6.6	X	X
Load losses	3.9.11, 3.9.5	4.6.7	X	X
Dielectric	3.23	4.6.8	X	X
Insulation resistance	3.24	4.6.8.4	X	X
Partial discharge	3.35	4.6.9	X	X
Temperature rise	3.8, 3.8.1	4.6.10	X	--
Short-circuit	3.9.8	4.6.11	X	--
Noise	3.13	4.6.12	X	X
Vibration	3.11	4.6.13	X	--
Shock	3.10	4.6.14	X	--
EMI	3.16	4.6.15	X	--
Degree of enclosure	3.25	4.6.16	X	--
Humidity	3.17	4.6.17	X	--
Inclined operation	3.18	4.6.18	X	--
Shipboard power	3.9.1	4.6.19	X	--

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TABLE III. Examinations and tests – Continued.

Test requirement	Requirement	Verification	First article	Conformance
Arc-resistant design	3.20	4.6.20	X	--
DC magnetic field	3.21	4.6.21	X	--
Inrush current	3.9.14	4.6.22	X	--
Insulation sample	3.4.4.1	4.6.23	X	--
Temperature storage	3.7.1	4.6.24	X	--
Measurement of frequency response	3.31	4.6.25	X	X
Toxicity	3.32	4.5.2	X	--
Prohibited materials	3.33	4.5.3	X	--
Off-gassing	3.34	4.7	X	--
Voltage regulation	3.36	4.6.26	X	X

NOTE:
1/ Indicates a requirement that may be verified with appropriate documentary evidence.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.5 and the tests of 4.6.2 through 4.6.23 (see table III). Conformance inspection shall be performed on each unit.

4.4 Inspection conditions. Unless otherwise specified herein (see 6.2), the examination and tests of 4.5 and 4.6 shall be performed under the following conditions:

a. Ambient temperature from 32 to 122 °F (0 to 50 °C)

b. Attitude: Normal operation position

Unless otherwise specified (see 6.2), ambient conditions within the specified range need not be controlled.

4.5 Examination.

4.5.1 General examination. The transformer shall be subjected to a thorough examination to determine that the material, workmanship, safety to operating personnel, design, and construction are in conformance with this specification. Examination shall include, but need not be limited to, the following:

a. Electrical connections and wiring in conformance with applicable documents and a continuity (ring out) of wired connections. Ground connections shall be inspected for resistance in accordance with MIL-STD-1310.

b. Outline and mounting dimensions in conformance with applicable documents (not needed for conformance testing).

c. Designation and marking, including safety warnings, are as specified in the applicable documents.

d. Weights and centers of gravity are in conformance with applicable documents (not needed for conformance testing). Centers of gravity may be determined by actual measurement or through the use of three-dimensional models.

e. Check of workmanship and fastener integrity; check for damage to platings or components.

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f. Operational test shall be conducted by energizing control voltage and operating the equipment including space heaters and any cooling systems to verify all functional features.

g. The 100-hour burn-in test shall be conducted by energizing control voltage to the equipment for 100 hours at nominal voltage and frequency, ambient temperature, and with all inputs and outputs connected to effect maximum rated loading. The 100-hour burn-in test may be accomplished at the end of first article testing.

h. The required creepage and clearance distances shall be verified (not needed for conformance testing).

i. Verification of arc fault equipment functionality (as applicable).

4.5.2 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.32 as specified (see 6.2). The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the transformer using data provided by the manufacturer/distributor to the NMCPHC (see 3.32 and 6.4).

4.5.3 Prohibited materials. Prohibited materials shall be verified as specified (see 3.33 and 6.2).

4.5.4 Off-gassing. The transformer shall be evaluated for off-gassing in accordance with S9510-AB-ATM-010 chapter titled "Material Control Program" (see 3.34 and 6.5). If the Navy determines that off-gas testing is required, testing shall be conducted at a test facility approved by the Technical Authority (see 3.34). The Navy will review the off-gas test results and assign a usage category. Additionally, the Navy will assign a usage category if an administrative review is conducted in lieu of off-gas testing (see 3.34).

4.6 Tests.

4.6.1 Test equipment. Test equipment shall be as specified in each individual test herein. Test equipment requiring calibration shall be certified to be properly calibrated at the time of testing.

4.6.2 Maintainability. Compliance shall be verified through a maintenance demonstration procedure, maintenance task selection, and maintenance task performance. For guidelines, refer to test method 1 of MIL-HDBK-470.

4.6.3 Resistance measurement tests. Resistance measurement tests shall be performed on all transformer windings in accordance with IEEE C57.12.91. Winding resistance will vary from transformer to transformer; therefore, the acceptance criteria including tolerance shall be as specified by the vendor. As specified (see 6.2), test results obtained shall be recorded in the test report and shall become baseline data for use in calculations used in performing the temperature rise test (see 4.6.10).

4.6.4 Polarity and phase-relation tests. Polarity and phase-relation tests shall be performed in accordance with IEEE C57.12.91. Failure to meet the requirements of IEEE C57.12.91 and IEEE C57.12.01 shall be cause for rejection.

4.6.5 Ratio test. A ratio test shall be performed in accordance with IEEE C57.12.91. Failure to meet the requirements of IEEE C57.12.91 and IEEE C57.12.01 shall be cause for rejection.

4.6.6 No-load loss and excitation current test. A no-load loss test shall be performed in accordance with IEEE C57.12.91. Failure to meet the requirements of IEEE C57.12.91 and IEEE C57.12.01 shall be cause for rejection.

4.6.7 Load loss and impedance voltage test. A load loss and impedance voltage test (see 6.3.3) shall be performed in accordance with IEEE C57.12.91. Procedures for transformers with more than three windings shall require NAVSEA approval. Failure to meet the requirements of 3.9.5 shall be cause for rejection.

4.6.8 Dielectric tests. Dielectric tests shall be performed in accordance with IEEE C57.12.91 and as specified in 4.6.8.1 through 4.6.8.4 at the levels prescribed by MIL-STD-1399-300-1 and MIL-STD-1399-300-2. The applied voltage test and induced voltage test shall be the only dielectric tests required for conformance testing. All dielectric tests shall be performed with grounding stud boots removed.

4.6.8.1 Impulse test. An impulse test (BIL) shall be performed in accordance with IEEE C57.12.91 using the BIL voltage levels defined in table II. Failure to meet the requirements of IEEE C57.12.91 and IEEE C57.12.01 shall be cause for rejection.

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4.6.8.2 Applied voltage test. An applied voltage test shall be performed in accordance with IEEE C57.12.91. Failure to meet the requirements of IEEE C57.12.91 and IEEE C57.12.01 shall be cause for rejection.

4.6.8.2.1 Power frequency withstand test. A power frequency withstand test shall be performed in accordance with MIL-STD-202-301 for low voltage control circuits, such as space heaters, at levels specified in MIL-DTL-917.

4.6.8.3 Induced voltage test. An induced voltage test shall be performed in accordance with IEEE C57.12.91. Failure to meet the requirements of IEEE C57.12.91 and IEEE C57.12.01 shall be cause for rejection.

4.6.8.4 Insulation resistance test. An insulation resistance test for windings shall be performed in accordance with IEEE C57.12.91. For low voltage control circuits, such as space heaters, an insulation resistance test shall be conducted in accordance with MIL-STD-202-302.

4.6.8.4.1 Test setup. The test setup for control circuits shall be as follows:

a. Test potential: 500 volts ±10 percent for operating potentials of 450 VAC or less.

b. Points of measure: Between each electrically isolated circuit and all other circuits connected together to ground (frame, chassis, or enclosure, as applicable).

c. Application time: The application time of test voltage shall be sufficient to take resistance readings.

d. Temperature at time of test: The temperature of parts to be tested shall be measured and recorded.

4.6.8.4.2 Acceptance criteria. For windings, failure to meet the requirements of IEEE C57.12.91 and IEEE C57.12.01 shall be cause for rejection. In addition, insulation resistance measurements shall be corrected to 77 °F (25 °C). For low voltage control circuits, insulation resistance shall be not less than 10 megohms corrected to 77 °F (25 °C) for any readings. Correction shall be made on the basis of insulation resistance doubling for each 27 °F (15 °C) decrease in temperature.

4.6.9 Partial discharge test. A partial discharge test shall be performed in accordance with IEEE C57.12.01, except for the following modifications:

a. Partial discharge inception/extinction voltage shall be greater than 1.3 Vr, where Vr is the rated line-to-line voltage. A lower value may be used if approved by the Technical Authority.

b. If 10 picocoulombs (pC) is beyond the capability of the equipment design and testing facility, a higher acceptable partial discharge intensity (greater than 10 pC) may be used if approved by the Technical Authority. Failure to meet the requirements of IEEE C57.12.91 and IEEE C57.12.01 shall be cause for rejection.

4.6.10 Temperature rise test. A temperature rise test shall be performed in accordance with IEEE C57.12.91 with the exception that the ambient temperature used for the test shall be 122±9 °F (50±5 °C).

4.6.10.1 Thermal imaging windows. When required (see 6.2), thermal imaging windows shall be tested to verify the capability to view chosen current carrying joints or expected hot spots.

4.6.10.2 Space heater test. The space heater test shall occur when the transformer is de-energized. Using an ambient temperature of 122±9 °F (50±5 °C), the heater circuit shall demonstrate that the heaters will increase the air temperature within the transformer enclosure over a time period until temperature stabilization in accordance with IEEE C57.12.91 is achieved.

4.6.10.3 RTD test. RTD functionality and accuracy as compared to test instrumentation shall be documented during the temperature rise test using instrumentation as described in 3.8.1.

4.6.10.4 Acceptance criteria. The transformer shall be capable of operating at full rated kVA in an ambient temperature of 122±9 °F (50±5 °C). The maximum temperature rises shall not exceed the temperature limits in 3.8. In addition, any subassembly component with a temperature rating shall also be verified to be operating within its rated temperature. During testing, the temperatures shall not exceed temperature limits with the space heater operational. Moisture shall not be visually observed to condense on the transformer internals with the space heater operational.

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4.6.11 Short-circuit test. A short circuit test shall be performed in accordance with IEEE C57.12.91 using the applied levels in 3.9.8. Failure to meet the requirements of IEEE C57.12.91 and IEEE C57.12.01 shall be cause for rejection. In addition, there shall be no evidence of shifting of coils or cores, damage to the insulation, or loosening of the coil supports.

4.6.12 Noise. The transformer assembly shall be tested for airborne and structureborne noise in accordance with MIL-STD-1474 and MIL-STD-740-2, respectively, to determine compliance with 3.13.

4.6.13 Vibration. The transformer shall be subjected to the type I tests of MIL-STD-167-1. Unless otherwise specified (see 6.2), the maximum frequency of vibration shall be in accordance with MIL-STD-167-1. The test setup and acceptance criteria shall be the same as specified in 4.6.14.1 and 4.6.14.2.

4.6.14 Shock. The transformer (principal unit) designed for hard mounting shall be subjected to grade A, class I, type A, high-impact shock testing in accordance with MIL-DTL-901. The transformer (principal unit) designed for shock isolators shall be subjected to grade A, class II, type A, high-impact shock testing in accordance with MIL-DTL-901.

4.6.14.1 Shock test setup. Shock test setup shall include the following:

a. All equipment shall be installed in accordance with the shipboard configuration during testing.

b. All lengths of ship's cabling and buswork shall be installed in the enclosure in accordance with the shipboard configuration during testing.

c. All lengths of ship's cabling that extend outside the foundation of the transformer shall be stubbed to 12 to 18 inches (30.5 to 46 centimeters) with the bitter ends insulated.

d. Cable, cable terminations, and cable restraints required for shipboard configuration shall be included in shock testing.

e. The test shall be conducted with all equipment energized.

f. Mounting location, mounting plane, and mounting orientation for the shock test shall simulate the worst-case shipboard installation, as specified (see 6.2).

g. Fans and water cooling, if applicable, shall be energized during this test.

h. Foundations used in shock testing shall simulate shipboard configuration.

i. The transformer shall be instrumented with a sufficient number of accelerometers to support shock analysis of components.

4.6.14.2 Acceptance criteria. Acceptance criteria shall be in accordance with MIL-DTL-901 and the following:

a. Mechanical inspection.

(1) Hairline cracks are permitted in the framework, but no separation of parts shall result.

(2) Any resulting distortion shall not be of sufficient magnitude to impair the operation of the equipment contained within.

(3) The enclosure shall not distort to an extent that might result in contact with any electrically conducting part within the transformer.

(4) The bus work shall not be mechanically damaged, nor the bus clearance distances reduced below the minimum requirements.

(5) When applicable, there shall be no observable leakage of fluids.

b. Electrical tests. After the mechanical inspection, the transformer shall be subjected to the following electrical tests in the order listed:

(1) Insulation resistance test in 4.6.8.4.

(2) Operations test in 4.5.1.f.

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4.6.15 EMI tests. Unless otherwise specified (see 6.2), the EMI tests of MIL-STD-461 (both below deck on surface ships as well as installation aboard submarines) shall be conducted on transformers with electronic devices. Testing shall not cause failure of electronic devices. Fans, if applicable, shall be energized during EMI testing.

4.6.16 Degree of enclosure. Unless otherwise specified (see 6.2), the transformer shall be subjected to the test requirements of MIL-STD-108 for a drip-proof 15-degree protected enclosure except that visual examination shall not be permissible to determine acceptability of drip-proof design. Fans, if applicable, shall be energized during this test.

4.6.16.1 Pass/fail criteria. Within 30 minutes of the completion of testing, the insulation resistance tests (see 4.6.8.4) shall be successfully conducted. There shall be no accumulation of liquid water within the enclosure.

4.6.17 Humidity test. The transformer shall be tested in accordance with MIL-STD-202-103 for a minimum of 96 hours to determine compliance with humidity requirements. The transformer shall be operational during the tests. Fans, if applicable, shall be energized during this test.

4.6.17.1 Pass/fail criteria. Within 30 minutes of the completion of testing, the insulation resistance tests (see 4.6.8.4) shall be successfully conducted. There shall be no accumulation of liquid water within the enclosure.

4.6.18 Inclined operation. Unless otherwise specified (see 6.2), the transformer shall not be damaged or caused to malfunction when subjected to the interface requirements and constraints of DOD-STD-1399-301 using the design limits for surface ships with sea state 8 or submarines.

4.6.19 Shipboard power test. If applicable, the low voltage power lines of the transformer using shipboard power shall be subjected to the complete series of tests in accordance with MIL-STD-1399-300-1, type I power. The medium voltage power lines of the transformer using shipboard power shall be subjected to the complete series of tests in accordance with MIL-STD-1399-300-2. Testing shall not cause false signals or failure of electronic devices.

4.6.20 Arc-resistant design (pressure relief). When specified (see 6.2), testing of pressure-relief features shall be conducted in accordance with IEEE C37.20.7. The fault levels that will occur at locations in the transformer and the permissible venting locations and clearances shall be as specified by the vendor.

4.6.21 DC magnetic field test. The transformer with electronic devices shall be subjected to a DC magnetic field test in accordance with MIL-STD-1399-407. The transformer with electronic devices shall remain operational during the tests.

4.6.22 Inrush current test. Oscillograph measurement of the inrush current to the transformer shall be obtained with rated voltage at rated frequency applied to the primary windings to determine conformance with 3.9.14. The secondary windings shall be open-circuited. The MIL-STD-1399-300-2 procedure and acceptance criteria shall be used in conducting this test.

4.6.23 Insulation sample test. Samples of conductor insulating coverings such as bus bar insulation and boots shall withstand the maximum rated line-to-line voltage between the conductor and the outside surface of the insulating covering for a period of 1 minute (foil test) in accordance with IEEE C37.20.2 and MIL-STD-202-301.

4.6.23.1 Flame test. When required (see 3.4.1.4.3), flame testing of insulation material shall be performed in accordance with IEEE C37.20.2.

4.6.23.1.1 Extension of flame test. Requests for flame test extension shall include information from previous testing performed in accordance with 4.6.23.1.

4.6.24 Temperature storage test. The non-operational transformer shall be subjected to a high/low temperature test in accordance with MIL-STD-810, method 501, procedure I and method 502, procedure I. Acceptance criteria shall be successful completion of an operational test (see 4.5.1.f) after the temperature test.

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4.6.25 Measurement of frequency response. Frequency response measurement testing of the transformer shall be performed in accordance with guidelines established in IEC 60076-18 for information purposes only to establish a baseline for the transformer. This test shall be performed before and after short circuit testing. This test shall be performed again before and after shock and vibration testing.

4.6.26 Voltage regulation test. The transformer shall be subjected to a voltage regulation test to determine the relative linearity of the voltage regulation band throughout its entire load range in accordance with IEEE C57.12.91. The tolerance and acceptance criteria shall be as specified (see 6.2).

4.7 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is acceptable to the Technical Authority. Accreditation shall be obtained from a recognized accreditation body such as the American Association for Laboratory Accreditation (A2LA) (www.a2la.org) or the International Code Council's International Accreditation Services (IAS) (www.iasonline.org). The scope of accreditation shall include specific flammability and fire tests. All other fire test provisions shall be as specified (see 6.2 and 6.6).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The medium voltage transformer will be used in shipboard electrical power distribution systems to step-up or step-down the incoming voltage to provide the necessary power to the ship service and propulsion loads.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article inspection is required (see 3.1).
b. General requirements, if other than specified (see 3.2).
c. Safety requirements, if other than specified (see 3.3).
d. Materials and coatings, if other than specified (see 3.4.1).
e. Non-preferred component materials allowed (see 3.4.1.1).
f. Non-preferred component coatings allowed (see 3.4.1.3.2).
g. Power cable connections (see 3.4.3.2).
h. Enclosures, if other than specified (see 3.4.5).
i. Enclosure access panels and covers, if other than specified (see 3.4.5.2).
j. When the heater is thermostatically controlled (see 3.5).
k. Identification plates, if other than specified (see 3.6).
l. Additional temperature monitoring locations (see 3.7.2).
m. When temperature warning and shutdown signals are required (see 3.7.2.1).
n. Temperature measurement method, if other than specified (see 3.8.1).
o. Transformer rating requirements (see 3.9).
p. Standard power interfaces requirements, if other than specified (see 3.9.1).

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q. The level specified for kVA capacity (see 3.9.3).
r. Duty requirements, if other than specified (see 3.9.4).
s. Impedance (see 3.9.5).
t. Ratio level (see 3.9.6).
u. Polarity and phase relation requirements (see 3.9.7).
v. The short circuit rating, if other than specified (see 3.9.8).
w. Weight and volume envelope rating (see 3.9.9).
x. Excitation losses requirement, if other than specified (see 3.9.10).
y. Load losses requirement, if other than specified (see 3.9.11).
z. General schematic requirement (see 3.9.13).
aa. Inrush current (see 3.9.14).
bb. When a soft start circuit is used (see 3.9.14).
cc. When taps are required (see 3.9.15).
dd. Tap rating (see 3.9.15).
ee. When fans are required (see 3.12.1).
ff. Water-cooled systems (see 3.12.2).
gg. Water flow characteristics (see 3.12.2.2).
hh. When heat exchangers are required (see 3.12.2.3).
ii. Allowable airborne and structureborne noise level (see 3.13).
jj. EMI limits and criteria, if other than specified (see 3.16).
kk. Inclined operation application requirement and verification, if other than specified (see 3.18 and 4.6.18).
ll. When arc-resistant design (pressure relief) devices and testing are required (see 3.20 and 4.6.20).
mm. Maintainability requirement (see 3.22).
nn. Degree of enclosure requirement and testing, if other than specified (see 3.25 and 4.6.16).
oo. When thermal imaging windows are required (see 3.28 and 4.6.10.1).
pp. Viewing location requirements (see 3.28).
qq. When arc fault sensors are required (see 3.29).
rr. When medium voltage fuses are required (see 3.30).
ss. Requirements for an HHA (see 3.32 and 4.5.2).
tt. Requirement for prohibited materials verification (see 3.33 and 4.5.3).
uu. Voltage regulation requirements (see 3.36 and 4.6.26).
vv. Inspection conditions, if other than specified (see 4.4).
ww. Control of ambient conditions, if other than specified (see 4.4).
xx. Test report and baseline data requirements (see 4.6.3).
yy. Vibration test frequencies, if other than specified (see 4.6.13).
zz. Shock test setup requirements (see 4.6.14.1.f).
aaa. EMI testing, if other than specified (see 4.6.15).
bbb. Degree of enclosure, if other than specified (see 4.6.16).
ccc. Additional fire testing provisions (see 4.7 and 6.6).
ddd. Packaging requirements (see 5.1).

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6.3 Definitions.

6.3.1 Dry-type transformer. A dry-type transformer is a transformer cooled by the natural circulation of air.

6.3.2 Exciting current of a transformer. Current in any winding used to excite the transformer when all other windings are open-circuited; it is expressed in percent of the rated current of the winding in which it is measured.

6.3.3 Impedance voltage of a transformer. Impedance voltage of a transformer is the voltage required to circulate rated current through a winding of the transformer when another winding is short-circuited, with the respective windings connected for rated voltage operation; it is expressed in percent of the rated voltage of the winding in which the voltage is measured.

6.3.4 Primary winding. Primary winding is the winding on the input side.

6.3.5 Rated kVA of a transformer. Rated kVA of a transformer is the output that can be delivered for the time specified at approximately rated secondary voltage and rated frequency without exceeding the specified temperature limitations.

6.3.6 Ratio of a transformer. Ratio of a transformer is the turn ratio of the transformer.

6.3.7 Regulation of a transformer. Regulation of a transformer is the change in secondary voltage, expressed in percent of rated secondary voltage, which occurs when kVA output at a specified power factor is reduced from rated value to zero, with the primary impressed terminal voltage maintained constant.

6.3.8 Secondary winding. Secondary winding is the winding on the output side.

6.3.9 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.3.10 Transformer. A transformer is an electric device without continuously moving parts that by electromagnetic induction transfers electric energy from one or more circuits to one or more other circuits at the same frequency, usually with changed values of voltage and current. It typically includes an enclosure or multiple adjoining enclosures containing a transformer and associated devices such as temperature sensors, space heaters, fuses, and indicator lights.

6.3.11 Transformer tap. A transformer tap is a connection brought out of a winding at some point between its extremities, usually to permit changing the voltage ratio.

6.4 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, https://www.med.navy.mil/Navy-Marine-Corps-Public-Health-Center/.

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6.5 Material certification. Materials to be installed in submarines are to be controlled to prevent off-gassing, which contaminates the submarine's atmosphere and can result in health hazards to personnel or deleterious effects on machinery. These controls are administered through the Submarine Material Control Program, which is described in the Nuclear Powered Submarine Atmosphere Control Manual, S9510-AB-ATM-010 chapter titled "Material Control Program." Under the Submarine Material Control Program, all materials considered for use on submarines require certification and assignment of a usage category. Under the certification process, candidate materials are selected by Navy activities or contractors, and a request for certification is submitted to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. The certification request is accompanied by detailed information, including descriptions of the material, method of application, usage, and storage. A chemical analysis is conducted, which can be accomplished through off-gas testing. If off-gas testing is required, it must be conducted in a Government approved laboratory. Information pertaining to this test requirement may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. Based on the chemical analysis results, a usage category is assigned to the material defining whether, and to what extent, the material may be used on submarines.

6.6 Additional fire testing provisions. The Technical Authority reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements. A2LA (www.a2la.org) or International Code Council's IAS (www.iasonline.org) maintain a full list of the approved laboratories that are accredited to ISO/IEC 17025 and provide details on how to obtain and maintain accreditation (see 6.2).

6.7 Subject term (key word) listing.

Arc fault sensor
Bus
Bus bar
Bus connection
Cable
Coil
Core
Enclosure
Fan
Fuse
Ground
Heat exchanger
Insulation
Primary winding
Resistance temperature detector
Secondary winding
Space heater
Tap
Wiring

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CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – SH
Air Force – 99
DLA – CC

Review activities:
Army – CR4, CR, GL, MI
Navy – AS, CG
Air Force – 03, 19, 50, 84, 85
DLA – GS, GS1

Civil agency:
GSA – FAS

Preparing activity:
Navy – SH
(Project 6150-2023-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

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